Delay–Doppler Synchronization for Distributed Uplink Interference Nulling in LEO Networks

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***Abstract*—Distributed satellite receivers must compensate the relative delay and Doppler of a common interferer before coherent cancellation. We analyze two-branch interference nulling with finite training and derive a closed-form approximation to the mean residual interference-to-noise ratio (INR). The approximation separates wrong acquisition, local synchronization errors, and weight-learning losses, linking performance to input INR, branch power imbalance, training length, and effective search size. Under a Gaussian surrogate with small errors and efficient, approximately unbiased refinement, correct acquisition yields a high-INR plateau of $2/N_v$, where N_v is the valid training length. Synchronization and weight learning contribute equally to this limit, whereas persistent alignment bias adds a penalty that grows with input power. Monte Carlo experiments compare dense FFT/zoom and coarse FFT/gradient acquisition using matched final FIR refinement, branch-noise normalization, and independent test blocks. Both approach the predicted plateau at high input INR; coarse search reduces measured search time by $6.7\times$ at an approximately 0.7-dB acquisition-threshold penalty. Controlled power-ratio sweeps show how branch imbalance delays reliable acquisition at fixed stronger-branch INR. A separate Starlink snapshot evaluation demonstrates comparable nulling performance under changing satellite geometry.**

***Index Terms*—Satellite communications, interference nulling, delay–Doppler estimation, sample covariance, residual INR.**

I. Introduction

Low-Earth-orbit (LEO) constellations extend wireless coverage [1], [2], but their uplinks face ground-based interference and jamming [3]. Beamforming guided by radio environment maps [4] and null shaping under location uncertainty [5] mitigate interference using an onboard array. These methods address spatial uncertainty rather than alignment across satellites; small angular separation between a jammer and the desired transmitter can still limit rejection by one array.

Satellite cooperation offers additional spatial diversity. Distributed downlink beamforming has been analyzed with timing and Doppler mismatch [6], without addressing uplink interferer alignment. Our previous work [7] demonstrated achievable-rate benefits of distributed uplink anti-jamming through game-theoretic beamforming. Its equalized per-satellite model left distributed synchronization open. This paper follows that work by isolating interference alignment and quantifying the resulting nulling performance. Figure 1 shows the two cooperating branches; the desired transmitter provides context only.

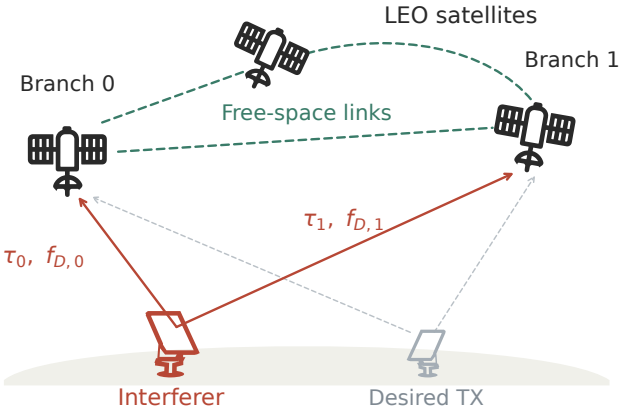


Fig. 1. Cooperative LEO interference reception. Red paths carry the common interferer; green links enable cooperation. Pale desired uplinks provide context; the model includes interference and noise.

Different path lengths and velocities create relative delay and Doppler. Terminal-side compensation supports massive-MIMO transmission [8]; cooperative downlink beamforming and timing-advance designs provide robustness to delay errors [9]. These designs synchronize communication links, leaving alignment of a separate ground interferer unresolved. Synchronizing to the desired user generally leaves relative offsets on interference paths, degrading coherent cancellation.

Generalized cross-correlation estimates delay between noisy replicas [10]; cross-ambiguity processing adds Doppler estimation [11]. The Keystone transform reduces wideband estimation cost [12]. These methods target parameter estimation, without analyzing subsequent finite-training nulling. Cramér–Rao bounds for unknown waveforms characterize local estimation precision [13], but do not quantify wrong-peak acquisition. Delay-error bounds capture ambiguity-driven threshold effects [14], without propagating them to cancellation performance.

Spatial filtering [15] and classical finite-sample adaptive-array analysis [16] characterize beamforming and weight-learning losses, without jointly modeling delay–Doppler acquisition. Covariance-based coexistence nulling [17] relates

spatial estimation to residual leakage, but uses local array snapshots without distributed waveform alignment. The remaining gap is a mean residual interference-to-noise ratio (INR) analysis combining wrong acquisition, local synchronization errors, and finite-training weight learning.

We study a common interferer and thermal noise at two receive branches. Interferer alignment need not align the desired replicas: desired-user timing/carrier synchronization, channel estimation, and signal-preserving combining remain future work, with equalization needed when intersymbol interference is present [18]. Residual INR alone does not establish desired-signal preservation or achievable rate.

Our contributions are as follows:

- **Closed-form residual-INR analysis:** We derive an approximation separating wrong acquisition, local synchronization, and finite-training weight-learning losses. It connects input INR, branch power ratio, training length, and effective search size, predicts the $2/N_v$ high-INR plateau under stated assumptions, and quantifies fixed-bias penalties.
- **Receiver comparison and acquisition–cost tradeoff:** We compare dense FFT/zoom and coarse FFT/gradient acquisition with matched final FIR refinement, noise-normalized covariance nulling, and independent test blocks. Coarse search reduces measured search time while retaining comparable high-INR nulling, at a higher acquisition threshold.
- **Power-imbalance and constellation evaluation:** Controlled INR and power-ratio sweeps characterize the weaker branch's effect on acquisition and assess the approximation's accuracy. A separate 24-hour Starlink snapshot study evaluates nulling under changing geometry and specified satellite-pairing constraints.

II. System Model and Receiver

A. Two Observations of a Common Interferer

The two satellite branches in Fig. 1 observe one common interferer and independent thermal noise. A unit-norm maximum-ratio combiner at an L-element array with single-path channel $\boldsymbol{a}_k = \alpha_k \boldsymbol{s}_k$, $\|\boldsymbol{s}_k\|^2 = L$, gives scalar gain $|g_k|^2 = L|\alpha_k|^2$ while preserving element-noise power N_0. Define the input INRs and power ratio by

$$\gamma_k = \frac{P_{\mathrm{tx}}|g_k|^2}{N_0}, \qquad \eta = \frac{\gamma_1}{\gamma_0} \le 1, \qquad \Gamma = \gamma_0 + \gamma_1. \tag{1}$$

After noise normalization, the observations are

$$y_k[n] = \sqrt{\gamma_k} e^{j\theta_k + j2\pi f_k n} x[n - \bar{\tau}_k] + d_k[n], \quad k \in \{0,1\}, \tag{2}$$

where x has unit average power, $d_k[n] \sim \mathcal{CN}(0,1)$, $\theta_0 = 0$, and θ_1 is an unknown constant phase. Fractional arguments denote interpolation. For sample interval T_{w}, $\bar{\tau}_k = \tau_k / T_{\mathrm{w}}$ and $f_k = f_{D,k} T_{\mathrm{w}}$; the relative parameters are $\Delta = \bar{\tau}_1 - \bar{\tau}_0$ samples and $\nu = f_1 - f_0$ cycles/sample ($N\nu$ frequency bins for N samples). Both absolute delays are retained in simulation because the shaped QPSK waveform is cyclostationary. The analysis uses a stationary surrogate.

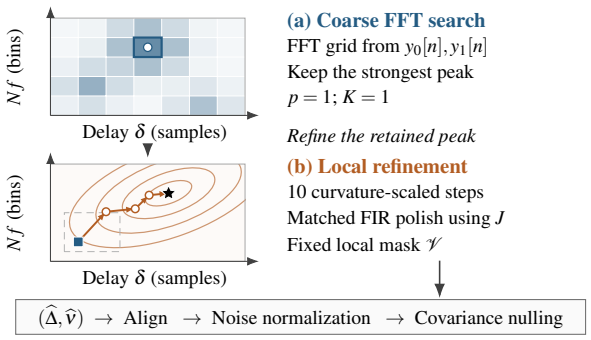


Fig. 2. Coarse acquisition and refinement ($p = 1$, $K = 1$), followed by alignment, branch-noise normalization, and nulling. The landscape and iterates are schematic.

B. Acquisition, Alignment, and Nulling

For a delay–Doppler candidate (δ, f), compensate branch 1 as

$$\widetilde{y}_1[n; \delta, f] = e^{-j2\pi f(n+\delta)} y_1[n + \delta]. \tag{3}$$

At (Δ, ν), the two interference copies differ only by a constant complex gain, absorbed by the nuller. An unnormalized FFT cross-ambiguity search initializes either dense zoom or coarse gradient refinement. Both then maximize the same normalized score over a local box $\mathcal{B}$:

$$J(\delta, f) = \frac{\left|\sum_{n\in\mathcal{V}} y_0[n] \widetilde{y}_1^*[n; \delta, f]\right|^2}{\sum_{n\in\mathcal{V}} |y_0[n]|^2 \sum_{n\in\mathcal{V}} |\widetilde{y}_1[n; \delta, f]|^2}. \tag{4}$$

The valid mask $\mathcal{V}$ excludes interpolation transients and nonoverlap and is fixed within each local box. The same de-rotation/FIR operator is used for refinement and nulling; a bounded L-BFGS-B solver approximately maximizes J. Write $N_v = |\mathcal{V}|$. Figure 2 illustrates the coarse path: the strongest FFT peak seeds curvature-scaled gradient updates, followed by matched final FIR refinement. A local numerical maximum does not guarantee statistical efficiency. The analysis treats the overlap length as a common N_v, although overlap can depend on the selected delay.

Using the selected alignment, form $\boldsymbol{z}[n] = [y_0[n], \widetilde{y}_1[n]]^T$ and $\widehat{\boldsymbol{R}} = N_v^{-1} \sum_{n\in\mathcal{V}} \boldsymbol{z}[n]\boldsymbol{z}[n]^H$. For aligned noise covariance $\boldsymbol{R}_d$,

$$\widehat{\boldsymbol{w}} = \arg\min_{\boldsymbol{w}\neq 0} \frac{\boldsymbol{w}^H \widehat{\boldsymbol{R}} \boldsymbol{w}}{\boldsymbol{w}^H \boldsymbol{R}_d \boldsymbol{w}}, \qquad \widehat{\boldsymbol{w}}^H \boldsymbol{R}_d \widehat{\boldsymbol{w}} = 1. \tag{5}$$

This is the minimum-eigenvalue rule after branch-noise whitening. For FIR coefficients $h_\tau[\ell]$ and temporally white input noise, $\boldsymbol{R}_d = \mathrm{diag}(1, \sum_\ell |h_\tau[\ell]|^2)$; this normalizes branch powers without whitening temporal correlation. With interference covariance $\boldsymbol{S}$, the target metric is

$$\overline{\mathsf{INR}} = \mathbb{E}_{\mathrm{train}}[\widehat{\boldsymbol{w}}^H \boldsymbol{S} \widehat{\boldsymbol{w}}]. \tag{6}$$

Independent test blocks estimate this residual, avoiding the optimistic training minimum eigenvalue. No desired signal enters the covariance.

III. Residual Interference Analysis

Let $\mathcal{C}$ denote correct main-lobe acquisition and $P_w = \mathbb{P}(\mathcal{C}^c)$. Conditioning gives the exact identity

$$\overline{\mathsf{INR}} = P_w \overline{\mathsf{INR}}_w + (1 - P_w)\overline{\mathsf{INR}}_c. \tag{7}$$

We approximate its terms using equal branch-noise powers, Gaussian surrogate statistics, effective independence of training samples and search cells, small synchronization errors, and efficient, approximately unbiased refinement. Sample correlation, selected-cell conditioning, and training-data reuse limit the approximation.

A. Correct Acquisition: Two Residual Contributions

With aligned waveform coherence r, the interference covariance is

$$\boldsymbol{S} = \begin{bmatrix} \gamma_0 & \sqrt{\gamma_0\gamma_1}r^* \\ \sqrt{\gamma_0\gamma_1}r & \gamma_1 \end{bmatrix}. \tag{8}$$

Its eigenvalues are

$$s_\pm = \frac{\Gamma \pm \sqrt{\Gamma^2 - 4\gamma_0\gamma_1(1-|r|^2)}}{2}. \tag{9}$$

Perfect alignment makes $\boldsymbol{S}$ rank one. Expanding its smaller eigenvalue near $|r| = 1$ gives the population alignment loss

$$s_- \simeq \frac{2\gamma_0\gamma_1}{\Gamma}(1 - |r|). \tag{10}$$

The population covariance $\boldsymbol{R} = \boldsymbol{I} + \boldsymbol{S}$ has eigenpairs $(\lambda_\pm, \boldsymbol{v}_\pm)$, with $\lambda_\pm = 1 + s_\pm$. Finite training perturbs its minimum-eigenvalue eigenvector toward the dominant mode. Writing $\boldsymbol{E} = \widehat{\boldsymbol{R}} - \boldsymbol{R}$, first-order perturbation gives

$$\widehat{\boldsymbol{w}} \simeq \frac{\boldsymbol{v}_- + \epsilon \boldsymbol{v}_+}{\sqrt{1 + |\epsilon|^2}}, \qquad \epsilon = \frac{\boldsymbol{v}_+^H \boldsymbol{E} \boldsymbol{v}_-}{\lambda_- - \lambda_+}. \tag{11}$$

Substitution into the independent-test quadratic form yields

$$\widehat{\boldsymbol{w}}^H \boldsymbol{S} \widehat{\boldsymbol{w}} \simeq \frac{s_- + s_+|\epsilon|^2}{1 + |\epsilon|^2} \simeq s_- + (s_+ - s_-)|\epsilon|^2. \tag{12}$$

Independent Gaussian samples give $\mathbb{E}|\epsilon|^2 = \lambda_+\lambda_-/[N_v(\lambda_+ - \lambda_-)^2]$. Since $s_+ - s_- = \lambda_+ - \lambda_-$, the additional interference is

$$\mathsf{INR}_{\mathrm{learn}} \simeq \frac{\lambda_+\lambda_-}{N_v(\lambda_+ - \lambda_-)} \simeq \frac{1+\Gamma}{N_v\Gamma}, \tag{13}$$

requiring $N_v\Gamma^2/(1+\Gamma) \gg 1$. Weight selection biases the training minimum eigenvalue downward; (13) instead measures learned-weight leakage on independent tests, as required by (6).

For a contiguous valid window, the stationary, separable model gives

$$|r| \simeq |R_x(\delta\tau)| \left| \frac{1}{N_v} \sum_{n=0}^{N_v-1} e^{j2\pi\delta\nu(n-\bar{n})} \right|. \tag{14}$$

Here $R_x(0) = 1$ and $\bar{n} = (N_v - 1)/2$; constant phase drops out. Expanding both factors for small errors $(\delta\tau, \delta\nu)$ gives

$$1 - |r| \simeq \frac{(2\pi)^2}{2}(\sigma_f^2 \delta\tau^2 + V_t \delta\nu^2), \quad V_t = \frac{N_v^2 - 1}{12}, \tag{15}$$

where σ_f^2 is normalized spectral variance and V_t is the variance of the centered sample indices. Combining (10), (13), and (15) gives a useful form before imposing efficient estimation:

$$\overline{\mathsf{INR}}_c \simeq \frac{1+\Gamma}{N_v\Gamma} + \frac{\gamma_0\gamma_1}{\Gamma}(2\pi)^2(\sigma_f^2 m_\tau + V_t m_\nu), \tag{16}$$

where $m_\tau = \mathbb{E}[\delta\tau^2 \mid \mathcal{C}]$ and $m_\nu = \mathbb{E}[\delta\nu^2 \mid \mathcal{C}]$ include variance and squared bias. This form quantifies synchronization loss without assuming statistical efficiency. Define $C = \gamma_0\gamma_1/[(1+\gamma_0)(1+\gamma_1)]$. A centered, separable surrogate information model uses

$$I_\tau \simeq N_v(2\pi)^2\sigma_f^2 \frac{2C}{1-C}, \qquad I_\nu \simeq N_v(2\pi)^2 V_t \frac{2C}{1-C}. \tag{17}$$

Centering the time index accounts for the unknown constant phase in this model. Assuming error variances I_τ^{-1} and I_ν^{-1}, each parameter contributes $(1-C)/(4CN_v)$ to $\mathbb{E}[1-|r|]$. Since $\gamma_0\gamma_1(1-C)/C = 1 + \Gamma$, substitution into (10) and addition of (13) yield

$$\mathbb{E}[s_-] \simeq \frac{1+\Gamma}{N_v\Gamma}, \qquad \overline{\mathsf{INR}}_c \simeq \frac{2(1+\Gamma)}{N_v\Gamma}. \tag{18}$$

□

Alignment and weight learning each contribute half, giving the high-INR limit $2/N_v$. Equation (17) is not an exact finite-window CRB for correlated RRC-shaped QPSK. The approximation cannot be extrapolated to vanishing power or biased refinement. Cancellation of spectral and temporal variance factors requires both parameters to attain their surrogate information scales. Colored samples and training-data reuse can alter the penalties and their interaction.

B. Incorrect Acquisition and the Mean Residual

At a wrong cell, decorrelated replicas give $\boldsymbol{S} \simeq \mathrm{diag}(\gamma_0, \gamma_1)$. Put $a = 1 + \gamma_0$, $b = 1 + \gamma_1$. Approximate the unnormalized correlation by $s \sim \mathcal{CN}(0, N_v ab)$, so $u = |s|^2/(N_v ab) \sim \mathrm{Exp}(1)$. Their maximum $U_{\max}$ over M independent cells satisfies, for integer M and $u \geq 0$,

$$\begin{aligned} F_{U_{\max}}(u) &= (1 - e^{-u})^M, \\ \mathbb{E}[U_{\max}] &= \sum_{m=1}^{M} \frac{1}{m} \simeq \ln M + \gamma_{\mathrm{E}} \equiv q_M, \end{aligned} \tag{19}$$

with Euler's constant $\gamma_{\mathrm{E}} \simeq 0.577$. We use the smooth approximation q_M for an effective, possibly noninteger cell count. Replacing the selected off-diagonal covariance power by abq_M/N_v and neglecting diagonal fluctuations gives

$$\cos\vartheta = \frac{\gamma_0 - \gamma_1}{\sqrt{(\gamma_0 - \gamma_1)^2 + 4abq_M/N_v}}, \tag{20}$$

$$\overline{\mathsf{INR}}_w \simeq \gamma_1 + (\gamma_0 - \gamma_1)\frac{1 - \cos\vartheta}{2}. \tag{21}$$

The selected minimum eigenvector has branch powers $|\widehat{w}_0|^2 \simeq (1 - \cos\vartheta)/2$ and $|\widehat{w}_1|^2 \simeq (1 + \cos\vartheta)/2$; weighting the independent test powers by these quantities gives (21). The residual approaches γ_1 for weak mixing ($\cos\vartheta \to 1$) and $\Gamma/2$ for strong mixing ($\cos\vartheta \to 0$).

At the correct cell, the surrogate $s \sim \mathcal{CN}(N_v\sqrt{\gamma_0\gamma_1}, N_v ab)$ implies $2u \sim \chi'^2_2(2N_vC)$. Approximating $\sqrt{u} \simeq \sqrt{N_vC} + \mathcal{N}(0,1/2)$ and comparing with $\sqrt{q_M}$ yields

$$P_w \simeq \tfrac{1}{2}\operatorname{erfc}\big(\sqrt{N_vC} - \sqrt{q_M}\big). \tag{22}$$

Here M counts effective independent cells, not oversampled grid points. The replacement of a random maximum by q_M omits its fluctuations and inter-cell dependence. Within this approximation, the $P_w = 1/2$ transition satisfies $N_vC = q_M$. If both branch INRs are small, $C \simeq \eta\gamma_0^2$, so $\gamma_{0,\mathrm{tr}} \simeq \sqrt{q_M/(N_v\eta)}$. Thus longer training improves acquisition, a larger search region raises the threshold logarithmically through q_M, and power imbalance delays reliable synchronization even at unchanged stronger-branch INR. Combining (18), (21), and (22) gives

$$\begin{aligned}\overline{\mathsf{INR}}(\gamma_0) \simeq P_w\left[\gamma_1 + (\gamma_0-\gamma_1)\frac{1-\cos\vartheta}{2}\right] \\ + (1-P_w)\frac{2(1+\Gamma)}{N_v\Gamma}.\end{aligned} \tag{23}$$

□

With $\gamma_1 = \eta\gamma_0$, the expression connects input INR, power imbalance, training length, and search size. It predicts an acquisition transition followed by the $2/N_v$ plateau, subject to the stated surrogate assumptions.

C. Fixed Bias and Coarse Acquisition

Under the small-error model in (16), fixed delay and Doppler biases (b_τ, b_ν) add

$$\Delta\mathsf{INR}_{\mathrm{bias}} \simeq \frac{\gamma_0\gamma_1}{\Gamma}(2\pi)^2(\sigma_f^2 b_\tau^2 + V_t b_\nu^2). \tag{24}$$

At fixed η and nonzero biases, this penalty grows linearly with input power and can exceed the nominal $2/N_v$ plateau.

Coarse sampling of the ambiguity surface can lower the sampled main-lobe peak and increase the risk of wrong acquisition. After correct acquisition, matched final FIR refinement can reduce grid-quantization errors. Identical refinement settings do not guarantee equal estimation accuracy or comparable wrong-peak statistics.

IV. Numerical Results

A. Configuration and Matched Receivers

Sionna RT 2.2 [19] regenerates two line-of-sight array channels from the first archived Starlink record [20]. A fixed archived unit-norm beam transmits RRC-shaped QPSK (Table I). The sample rate is 585.9375 kHz and occupied bandwidth is 244.14 kHz; the nominal 200-MHz link-budget bandwidth sets the original noise power and $T_{\mathrm{FFT}} = 5.12$ μs, not the probe bandwidth. Absolute delays and symbol-clock phase are retained in 3816-sample periodic records with 1140-sample guards.

The effective cell count is fixed by the archived search region and waveform bandwidth. With delay limit D, normalized Doppler limit $\nu_{\max}$, and normalized occupied bandwidth $B_x = (1+\beta)/S$, the time–bandwidth estimate is

$$M \simeq (2D+1)B_x(2\nu_{\max}N + 1). \tag{25}$$

TABLE I
Fixed-geometry configuration (rounded where shown).

Parameter	Value
Elements per array	6×6
Sample interval T_{w}	1.7067 μs
Record / valid samples	$N = 1536$ / $N_v = 1222$
Power ratio / INRs at 50-dBm TX	0.380 / 8.98, 4.78 dB
Absolute delays (samples)	759.607 / 1026.077
Relative Doppler	-7.876 kHz
RRC roll-off / samples per symbol	$\beta = 0.25$ / $S = 3$
FIR interpolator	96-tap Kaiser sinc, shape 9
Delay / Doppler search range	± 600 samples / ± 20 kHz
Training trials / test blocks per trial	100–300 / 32

TABLE II
Acquisition settings and median search/refinement time.

Search	FFT p	Initial refinement	**Time (ms)**
Dense	4	6 zoom stages	**99.0**
Coarse	1	10 gradient steps	**14.7**

Coarse: $K = 1$ peak, gradient step factor $\mu = 1$.

The delay and Doppler cell widths are approximately $1/B_x$ samples and $1/N$ cycles/sample. The tabulated ranges give $M \simeq 5.30 \times 10^4$; zero padding increases grid density, not the physical cell count. Cell correlation limits the independence approximation.

The input sweep has 31 settings from -25 to 30 dB, sampled more finely near acquisition. Receivers share training/test waveforms within trials; resetting seed 95000 at each power provides common random numbers across powers. Residuals are averaged linearly over independent test blocks and training trials before conversion to dB. Pointwise 95% intervals use 2000 bootstrap resamples of trial means; acquisition intervals use Wilson's method. Correct acquisition requires $|\widehat{\Delta} - \Delta| \le S/2$ and $N|\widehat{\nu} - \nu| \le 0.5$.

Table II summarizes acquisition settings. Both receivers then apply bounded L-BFGS-B refinement in a ± 0.6-sample, ± 0.6-bin box, clipped to search limits, with at most 30 iterations; updates must not decrease J. They share the FIR operator, fixed local mask, and noise normalization. The oracle knows the synchronization parameters but still learns weights from finite training.

B. Acquisition, Residual INR, and Cost

Figure 3 separates acquisition failure from local estimation loss. At $\gamma_0 = -7$ dB, dense and coarse search have wrong-acquisition rates of 26.7% and 44.2%, and residual INRs of -14.17 and -12.61 dB, respectively. Coarse search reaches $P_w = 0.1$ about 0.7 dB later, interpolating the measured curves. Wilson intervals remain nonzero even at zero observed errors (upper endpoint 3.1% for 120 trials).

After reliable acquisition, both methods give approximately -27.25 dB residual INR at 8.98-dB input and -27.75 dB at 30-dB input, close to $10\log_{10}(2/1222) = -27.86$ dB. At -5 dB input, however, the dense residual is -19.55 dB versus the predicted -22.59 dB. Correlated, cyclostationary samples, finite-window selection, and imperfect local efficiency limit the surrogate's accuracy.

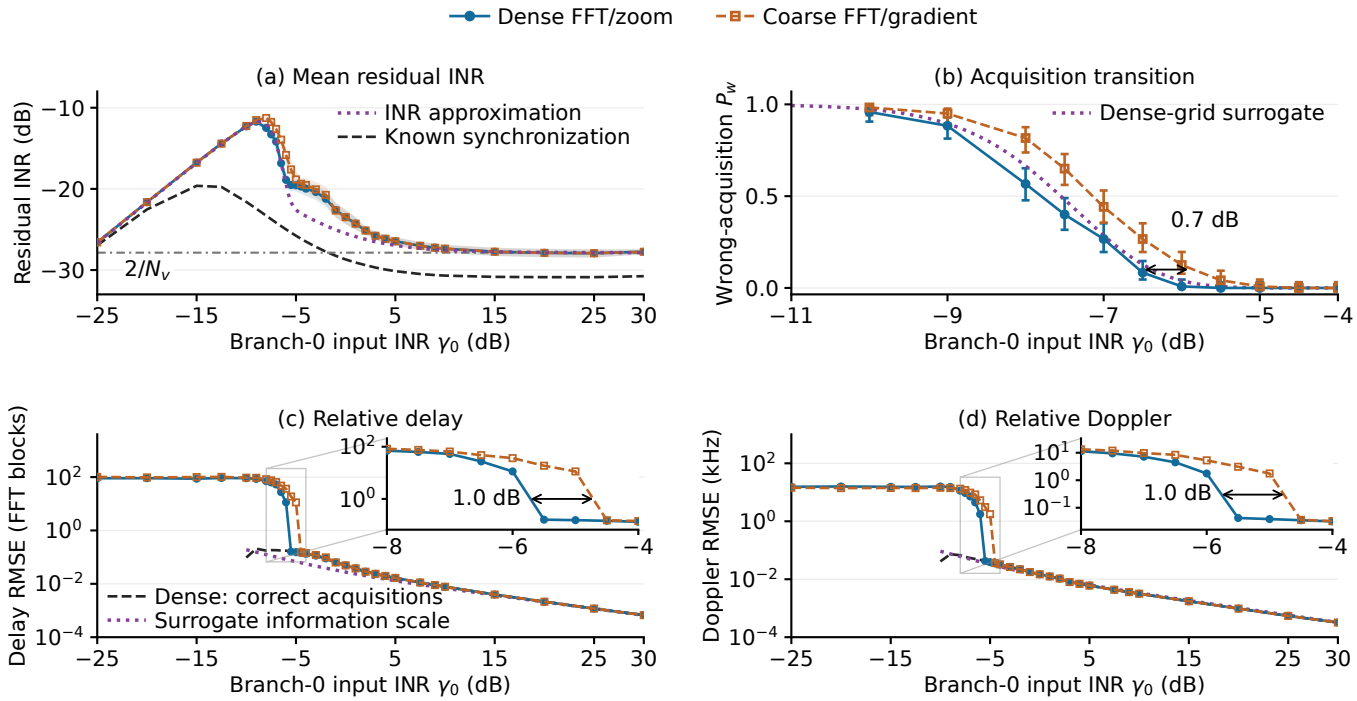


Fig. 3. Dense and coarse acquisition with matched FIR refinement: (a) mean residual INR and pointwise 95% bootstrap intervals; (b) wrong-acquisition rates and Wilson intervals; (c), (d) delay and Doppler RMSE, with transition zooms and equal-RMSE gap arrows. Black dashed: known synchronization in (a), correct dense acquisitions in (c), (d). Purple dotted: surrogate references; in (a), the dotted curve is (23), which reduces to (18) when acquisition failures are negligible. The gray horizontal line marks the high-INR limit $2/N_v$ of (18). The information scales in (c), (d) are not exact finite-window CRBs.

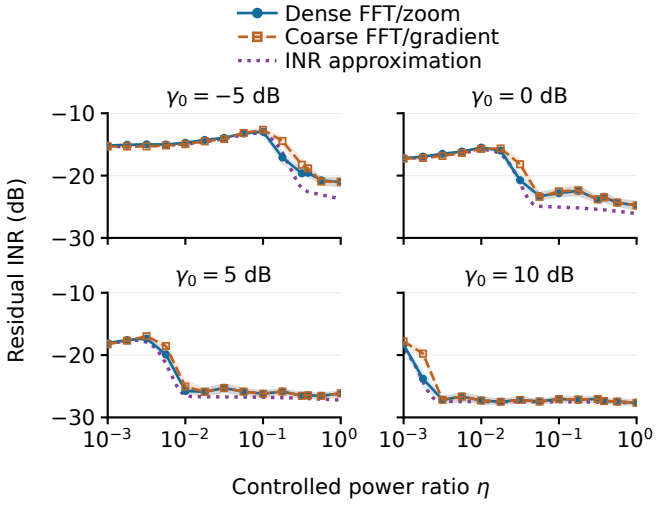


Fig. 4. Residual INR versus power ratio at four fixed input INRs. Shading: pointwise 95% trial-bootstrap intervals; dotted curves: (23) with $N_v = 1222$.

The RMSE panels show wrong-peak errors at low input INR and convergence toward the common information scale after reliable acquisition. Delay errors are divided by $S = 3$ for FFT-block units; Doppler errors are converted to kHz using T_w. Equation (17) is a model-dependent reference, not a strict bound for this QPSK experiment or for RMSE conditioned on successful acquisition.

On one Intel Core Ultra 9 275HX thread, 40 runs after two warmups at 8.98-dB input measure FFT search, candidate extraction, and all refinement (Table II), excluding channel generation, weight learning, and testing. Coarse search is 6.7× faster.

C. Controlled Power Imbalance

Figure 4 varies η over 13 logarithmic points from 10^{-3} to 1 plus the original ratio $\eta_0 \simeq 0.380$, for $\gamma_0 \in \{-5, 0, 5, 10\}$ dB. Scaling branch-1 gain by $\sqrt{\eta/\eta_0}$ preserves geometry, phases, delays, Dopplers, and noise. Each new point uses 120 paired trials and 32 independent tests; original-ratio anchors reuse 100–120 trials. The additional FFT searches are GPU-accelerated and checked against the CPU code; Table II retains the original CPU benchmark.

A stronger weak branch improves acquisition through C, but also increases the residual available after wrong acquisition. Thus a small residual at very small η need not imply successful synchronization. At $\gamma_0 = 0$ dB and $\eta = 0.03162$, dense/coarse methods have 8/120 and 26/120 wrong acquisitions and residuals of $-20.75/-18.18$ dB. At 10-dB input and $\eta = 1$, both give -27.65 dB, matching the surrogate. At -5 dB and $\eta = 1$, neither has an observed acquisition error, yet the dense residual is -21.04 dB versus -23.74 dB predicted: reliable acquisition alone does not ensure efficient refinement.

V. Long-Term Starlink Constellation Evaluation

A separate 24-hour experiment samples 720 orbital snapshots at 120-s intervals using the scalar LoS link budget in Table III. Among satellites with elevation at least 60°,

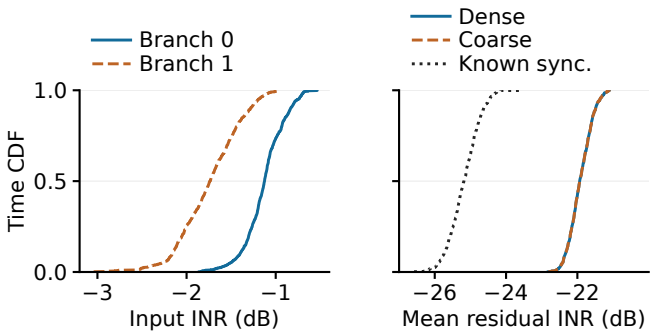


Fig. 5. Input and mean residual INR CDFs over 716 equally weighted Starlink snapshots. Dense and coarse residual curves nearly coincide.

TABLE III
STARLINK SNAPSHOT PARAMETERS (ROUNDED WHERE SHOWN).

Parameter	Value
Ground latitude / longitude	39.957° / -105.160°
Ground height / receive gain	1660 m / 0 dBi
Carrier frequency / EIRP	10 GHz / 50 dBm
Temperature / noise figure	290 K / 5 dB
Noise bandwidth / noise power	244.14 kHz / -115.10 dBm
Sample rate / samples per symbol	1.171875 MHz / $S = 6$
Record length / FIR length	$N = 2916$ / 192 taps
Delay / Doppler search range	± 0.30 ms / ± 125 kHz
Training trials / test blocks per trial	120 / 32

branch 0 is nearest; branch 1 is selected uniformly with a fixed seed from partners satisfying the tabulated relative delay and Doppler limits. This pairing rule retains 716 snapshots and keeps the dense FFT baseline tractable. Both methods search the same bounded region.

Bandlimited thermal noise is modeled in each branch-centered ideal receive passband, which preserves the full RRC spectrum. Absolute Doppler is retained; carrier centers are not supplied to the estimator. Output-noise normalization accounts for temporal correlation.

Both methods use matched final FIR refinement with at most 30 iterations in a ± 1.2-sample, ± 0.6-bin box around the initial estimate, clipped to the search limits. Per-snapshot residuals average the tabulated training trials and independent test blocks in linear power.

In Fig. 5, branch input INRs range over $[-1.86, -0.54]$ and $[-3.04, -0.97]$ dB. Both receivers achieve a median per-snapshot mean residual of -21.94 dB and a 95th percentile of -21.37 dB, with no observed wrong acquisition in 85,920 training trials. The known-synchronization median is -25.18 dB, revealing a remaining alignment penalty. These CDFs summarize per-snapshot means, not pooled trials; adjacent snapshots are temporally correlated. Different noise bandwidth and record length preclude testing the fixed-geometry $2/1222$ plateau here.

VI. CONCLUSION

The mean residual INR approximation connects acquisition, synchronization, and finite-training losses, predicting a $2/N_v$ plateau under stated assumptions and quantifying imbalance and bias effects. With matched refinement, coarse search approaches dense-search nulling after reliable acquisition while reducing measured search time by $6.7\times$ at a roughly 0.7-dB acquisition penalty. Starlink snapshots support comparable nulling under changing geometry. Desired-user synchronization, signal-preserving combining, and equalization where needed remain future work.

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